

KCNA - Kubernetes and Cloud Native Associate

- 1st Stage of k8s certification
- Foundation
- Exam - 1000 Marks - 750/1000 - Pass
- Cost - 250\$

Domains & Competencies

Kubernetes Fundamentals46%

Kubernetes Resources
Kubernetes Architecture
Kubernetes API
Containers
Scheduling

Container Orchestration22%

Container Orchestration Fundamentals
Runtime
Security
Networking
Service Mesh
Storage

Cloud Native Architecture16%

Autoscaling
Serverless
Community and Governance
Roles and Personas
Open Standards

Cloud Native Observability8%

Telemetry & Observability

Prometheus

Cost Management

Cloud Native Application Delivery8%

Application Delivery Fundamentals

GitOps

CI/CD

What is a Cloud-Native ?

- Cloud native refers to an approach to designing and building software applications that fully leverages the capabilities and advantages of cloud computing. It involves a shift from traditional, monolithic architectures to more modular, distributed, and scalable solutions. Cloud native applications are built with microservices, containerized with technologies like Docker, and orchestrated with platforms like Kubernetes.

Key characteristics of cloud native applications:

Microservices:

- Applications are broken down into smaller, independent, and loosely coupled services.

Containers:

- Applications are packaged in containers, like Docker containers, for portability and consistency across different environments.

Orchestration:

- Container orchestration platforms, like Kubernetes, manage and automate the deployment, scaling, and lifecycle of containerized applications.

Automation:

- Cloud native development and operations often rely on automation tools and practices to streamline workflows and improve efficiency.

Cloud-based:

- Cloud native applications are designed to run and scale seamlessly within a cloud environment, taking advantage of its elasticity and resources.

Scalability:

- Cloud native applications are designed to be highly scalable, allowing them to adapt to changing workloads and demands.

Resilience:

- Cloud native architectures are designed to be resilient, with multiple instances of services and automatic failover mechanisms.

Continuous Delivery:

- Cloud native development often embraces continuous delivery practices, allowing for frequent and rapid updates to applications.

Benefits of cloud native:

Increased Agility:

- Cloud native applications are more agile, allowing for faster development cycles and quicker responses to changing business needs.

Improved Scalability:

- Cloud native applications can scale up or down quickly and easily as needed, ensuring optimal performance and resource utilization.

Enhanced Resilience:

- Cloud native architectures are designed to be resilient, with multiple instances of services and automatic failover mechanisms.

Cost Optimization:

- Cloud native applications can be more cost-effective, as they leverage the on-demand resources and pay-as-you-go model of the cloud.

Simplified Management:

- Cloud native tools and practices can simplify application management, allowing for easier monitoring, logging, and troubleshooting.

Cloud Native V/s Cloud-Service-Provider

Cloud-Native:

Focus:

- - Building applications specifically designed for the cloud, leveraging its scalability, flexibility, and reliability.

Key Characteristics:

- - Microservices architecture

- - Containerization (using Docker)
- - Orchestration (using Kubernetes)
- - Continuous deployment and delivery

Benefits:

- Increased agility, scalability, and resilience
- Faster time-to-market
- Improved resource utilization

Example:

- A web app built with microservices, deployed in containers, and orchestrated by Kubernetes on a cloud platform.

Cloud Service Provider (CSP):

Focus:

- Providing on-demand cloud computing resources and services (IaaS, PaaS, SaaS).

Key Characteristics:

- Offers infrastructure, platforms, and software as services, managed by the CSP.

Examples:

- AWS, Azure, Google Cloud Platform, DigitalOcean, Rackspace.

Benefits:

- Reduced capital expenditures
- Increased flexibility and scalability
- Access to specialized services